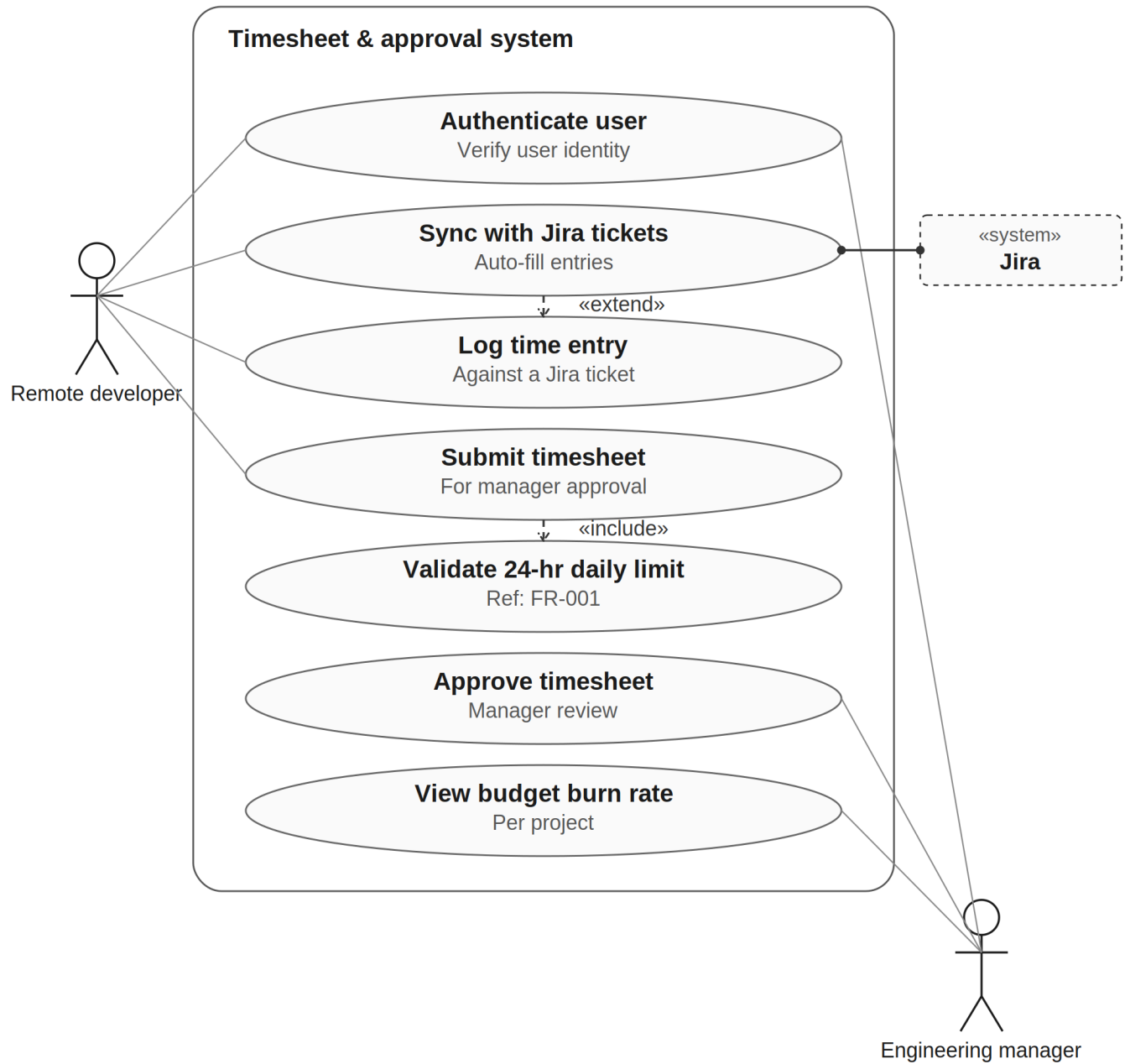


LAB 1 SRN: PES1UG24AM395 NAME: SAKSHI P SHETTY SEC: G

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow developers to log time against specific project tasks and prevent timesheet submissions exceeding 24 hours in a single calendar day.	High	Pass: Valid timesheet submitted for manager approval. Fail: Timesheet logging 30 hours in one day accepted.	Core ordering/logging functionality (given)
FR-002	Functional	The system shall allow a developer to submit a completed timesheet to their assigned Engineering Manager for approval.	High	When "Submit" is tapped with all daily entries filled, the timesheet status changes to "Pending approval" and the manager's queue updates within 2 seconds.	Enables the core approval workflow
FR-003	Functional	The system shall allow an Engineering Manager to approve or reject a submitted timesheet, with a required comment on rejection.	High	Pass: Rejecting without a comment is blocked with an inline error. Fail: Rejection saved with an empty comment field.	Managers need a documented reason for disputes and audits
FR-004	Functional	The system shall calculate and display each project's budget burn rate as a percentage of allocated hours consumed, updated whenever a timesheet is approved.	Medium	Pass: Burn-rate percentage recalculates and displays within 1 minute of a timesheet approval. Fail: Figure remains stale after approval.	Lets managers catch budget overruns before they become critical
FR-005	Functional	The system shall link each logged time entry to a valid, open Jira ticket ID and reject entries referencing a closed or nonexistent ticket.	Medium	Pass: Entry against a closed ticket is rejected with an error message. Fail: Entry saved against a closed or invalid ticket ID.	Keeps logged hours traceable to real, active work
NFR-001	Nonfunctional (Performance & Security)	Timesheet approval status transitions and project budget burn charts must update across dashboards within 500 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given requirement
NFR-002	Nonfunctional (Security)	The system shall enforce role-based access control so that a Remote Developer can only view/edit their own timesheets, and an Engineering Manager can only approve timesheets for developers on their own team.	High	Pass: A penetration test confirms a developer account cannot read or modify another developer's timesheet via API or UI. Fail: Cross-account access succeeds.	Protects sensitive salary-adjacent time and budget data

UML Use-Case Diagram

Scenario #46 - Remote Team Time-Tracking & Project Approver



Use-Case Flow Specification

Lab 1: Requirements Engineering & UML Use-Case Modelling | Scenario #46 - Remote Team Time-Tracking & Project Approver

Use case: Submit Timesheet

Primary actor:

Remote Developer

Preconditions:

The developer is logged in, and at least one time entry exists for the current period, with every entry linked to a valid, open Jira ticket.

Postconditions:

The timesheet's status is set to "Pending approval," the entries are locked from further editing, and the timesheet appears in the assigned Engineering Manager's approval queue.

Main success scenario:

1. Developer selects "Submit timesheet" for the current pay period.
2. System validates that no single day exceeds 24 logged hours and that every entry references an open Jira ticket (FR-001, FR-005).
3. System locks the submitted entries from further editing.
4. System sets the timesheet status to "Pending approval" and adds it to the Engineering Manager's queue.
5. System notifies the assigned Engineering Manager that a timesheet is awaiting review.
6. Use case ends successfully.

Alternate flow - 2a. Validation fails:

- 2a1. System flags the offending entry (a day exceeding 24 hours, or an entry against a closed/invalid Jira ticket) and displays a specific error message.
- 2a2. Developer edits the flagged entry to correct it.
- 2a3. Developer re-submits the timesheet; use case resumes at step 2.